

University of Bologna

Master in Artificial Intelligence

Fair and Constraint-Aware Pharmacy Shift Scheduling

A Computational Perspective on Fairness, Robustness, and Explainability

Ethics in Artificial Intelligence

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This report presents an auditable decision-support pipeline for pharmacy duty scheduling in the Bologna city area, based exclusively on the supplied calendar, with a focus on workload fairness, technical robustness, transparency, privacy, and human oversight.

Abstract

This project investigates pharmacy duty scheduling as a constrained allocation of workload. The sole empirical input is the supplied TURNI CSV. The Bologna city-area calendar contains 53 weekly blocks, 123 rotating pharmacies and 931 assignments, organized into 42 recurring patterns and seven dominant rotation groups. A mixed-integer linear program preserves a historical staffing floor, H24 duty count, primary-group eligibility, a horizon workload ceiling and a policy of balanced H24 cycle counts. It compares efficiency-first, fairness-aware and full-group scenarios using lexicographic priorities.

The selected fairness-aware schedule has 923 assignments, total workload range 1 and H24 range 1, compared with 931, 2 and 3 historically. Total Jain improves slightly, but total Gini increases slightly; fairness improvement is metric-specific. 11 complete nT patterns change, including 2 with H24-only changes.

The robustness experiment tests 738 eligible pharmacy-pattern cases, of which 730 affect scheduled pharmacies. 424 (58.1%) have a materialized, model-feasible same-group response. The 306 remaining scheduled cases (41.9%) require human intervention, "requiring human intervention" means that the model cannot restore the staffing floor within the absent pharmacy's primary group: a person would need to review a cross-group substitution or explicitly revise a modeled constraint. The algorithm does not make that decision. Of these cases, 17 have a historically evidenced, model-feasible candidate and 289 have no pre-filled candidate; none is automatically authorized.

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1 Original dataset and project overview

This project studies the pharmacy duty rotation in the original TURNI CSV. Its question is how to retain the observed recurring structure while comparing count-based workload fairness, testing hypothetical disruptions and explaining the resulting decisions. The analysis proceeds through six notebooks, each consuming saved evidence from earlier stages.

1.1 What the original file contains

The sole empirical source is a semi-structured spreadsheet export containing date blocks, nT pattern labels, shift-section headers, pharmacy names and addresses, and P/C access labels. Two facilities are explicitly marked as outside the rotation. The summary below reports quantities reconstructed from this original file by notebooks 00 and 01; it is not a second input dataset.

The source distinguishes H24 from STANDARD service, and P (open-door) from C (call/closed-door) access. The source headers describe H24 as Monday 08:30 to the following Monday 08:30, and STANDARD duty as 08:30-12:30 and 15:30-19:30, Monday through Sunday. These labels describe the source; the model counts weekly assignments rather than simulating actual opening hours. One assignment is a pharmacy-week duty block.

Table 1: Dataset reconstructed from the CSV.

Property	Value
Weekly blocks	53
Distinct nT patterns	42
Rotating pharmacies	123
Historical assignment rows	931
Historical H24 assignments	159
Historical STANDARD assignments	772
Weekly historical staffing	17-18
H24 pharmacies in each block	3
Maximum assignments per pharmacy in the horizon	8
Excluded non-rotation facilities	2
Group sizes, in group order	17, 18, 18, 17, 18, 18, 17

The first block begins on 29 December 2025 and ends on 5 January 2026. The final block begins on 28 December 2026 and is recorded as ending on 3 January 2027. That final end date is Sunday, six elapsed days after Monday start, inconsistent with a uniform Monday-to-Monday handover. Counting seven inclusive calendar dates does not resolve that timing inconsistency. The pipeline preserves the original boundary and records it as a source-data limitation; it does not guess a corrected date or require a second dataset to complete the analysis. Throughout this report, horizon workload means counts over all 53 source blocks, including the boundary anomaly, rather than hours or a strict 365-day total.

1.2 Scope of the analysis

The project uses only the CSV supplied in the original archive as empirical input. No geographic, actual-availability, demand, preference or external legal/contractual dataset is added.

The distinction between observation and modeling is maintained throughout. Counts, dates, identifiers, H24/STANDARD labels, P/C labels and historical co-occurrences are observations or deterministic derivations. Workload equality, scenario priorities, historical continuity and the choice to simulate one absence at a time are analytical design choices. They are justified explicitly rather than attributed to information absent from the file. References support mathematical methods and the ethics discussion; they provide no additional pharmacy data.

1.3 How the six notebooks connect

The six notebooks execute in order: cleaning and canonicalization; descriptive baseline; constraint specification; MILP optimization; pattern-level robustness; final comparison and reporting. They exchange explicit CSV and JSON artifacts rather than relying on notebook memory. All three optimized schedules are exported, allowing every scenario's reported metrics to be checked against its actual rows.

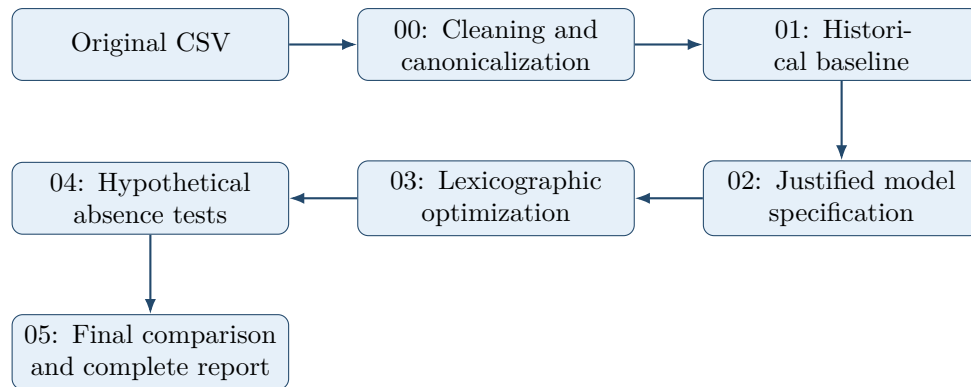


Figure 1: Dependency chain. All empirical inputs originate in the single supplied CSV; policies are specified before optimization.

The following chapters are in notebook order. Each chapter identifies its inputs, explains the computation, presents the final results and names the files passed downstream.

2 Notebook 00: cleaning and canonicalization

2.1 Purpose and inputs

Notebook 00 starts from the unchanged raw TURNI CSV and turns its section-based layout into one row per pharmacy-week assignment.

The supplied CSV is a semi-structured spreadsheet export with date blocks, shift-section headers and pharmacy assignment rows. Notebook 00 checks the expected column layout, normalizes Unicode and whitespace, accepts assignment rows with access mode P or C and excludes two rows explicitly marked as outside the duty rotation. Names and addresses are replaced by pharmacy and location identifiers; source-row positions are retained for traceability.

2.2 Pseudonymization and source fidelity

Sequential pseudonyms are not anonymization. Source-row positions, rare schedule features and the public calendar can enable linkage. The raw CSV and private mapping files contain the identifying information supplied with the project and are retained for reproducibility. Analytical outputs use pseudonyms, but are not claimed to be irreversibly anonymous. Existing mappings are validated and new IDs start above the largest existing identifier, preventing collisions when a later dataset adds pharmacies or locations.

The reconstructed calendar has 124 distinct pharmacy-location identifiers for 123 pharmacies. 118 pharmacies have mode P and 5 have mode C; no pharmacy changes P/C label across its observed rows. One pharmacy has two observed locations. Locations are retained as identifiers and are not converted into a geographic coverage model. Every accepted row preserves its source-row reference, and all accepted rows occur exactly once in the canonical output.

2.3 Reconstructing dates, patterns and rotation groups

The canonicalizer reconstructs dates, nT labels, groups, rounds, shift types and access modes. It verifies row counts, unique pharmacy-week assignments, consecutive weekly starts, adjacent boundaries, nT progression and consistent week metadata. The baseline checks repeated patterns and the constant H24 count. Identical nT values reuse the same pharmacy, location, shift and access signature across their calendar occurrences; there are 11 matching repeated-pattern comparisons out of 11.

For candidate periods 2 through 41, membership purity is the sum, across pharmacies, of the largest assignment count in any nT residue class, divided by all assignment rows. The leading period is 7 with purity 0.998926. One historical cross-group row means purity is not perfect. Dominant historical membership determines each pharmacy's primary modeled group; this is an inference from observations, not an externally supplied authorization list.

2.4 Exploratory permutation test

The revised permutation sensitivity test randomizes the 42 nT labels as intact pattern blocks, preserving each pattern's pharmacy membership and repeated-week multiplicity. Each random-

ization repeats the entire candidate-period search. It uses $B=500$ permutations and finds $b=0$ null maxima at least as extreme as the observed maximum. The corrected Monte Carlo p-value is $(b+1)/(B+1) = 0.001996$. Randomly sampled permutation p-values should not be reported as zero. [2]

This result supports the period under an exchangeable-pattern-label null. It is exploratory evidence, not proof of an administrative rule or stability in another year. The original row-level label shuffle disrupted within-pattern structure and repeated blocks; the revised procedure retains those dependencies. The significance claim is limited to this explicit randomization null.

Candidate periods are 2 through 41, inclusive. The random seed is 0. The mean of the null best-purity values is 0.665429, and their maximum is 0.794844. The number of permutations controls Monte Carlo resolution; it is a computational precision setting, not a pharmacy-data value or a fairness threshold. A 0.01 significance cutoff is an explicit screening choice used to stop the pipeline if this exploratory structural evidence is weak, rather than a requirement inferred from pharmacy data. The complete search is repeated for each draw, so the null comparison accounts for selection of the leading candidate.

2.5 Final reconstructed dataset

The notebook exports 931 accepted assignments across 53 source blocks, 123 pharmacy identifiers and 124 location identifiers. It retains the 42 pattern labels and infers seven dominant groups, with membership purity 0.998926. The two excluded facilities are recorded separately. The six-day final boundary is preserved as documented in the opening chapter. The canonical schedule is the sole schedule input to notebook 01; the observed profile preserves the raw and canonical hashes and the exploratory-test settings.

2.6 Final outputs and hand-off

`canonical_schedule.csv`, `pseudonymized_assignments.csv`, `observed_data_profile.json`, `mapping_PRIVATE.csv`, `excluded_non_rotation_PRIVATE.csv`, and `stage_00_manifest.json`. These files are supplied in `notebook_00/outputs/`. The executed notebook retains its displayed tables and figures.

3 Notebook 01: historical baseline

3.1 Purpose and inputs

Notebook 01 reads the canonical schedule from notebook 00 after checking its stage manifest. It measures the historical calendar before any optimization policy is applied. The baseline is descriptive: its extrema become possible policy values only when notebook 02 explicitly adopts and justifies them.

3.2 Workload and fairness definitions

One workload count is one pharmacy assigned to one source calendar block. Total and H24 counts are computed separately across all 123 pharmacies, retaining zero H24 counts.

For N nonnegative workloads w_i , define

$$\bar{w} = \frac{1}{N} \sum_{i=1}^N w_i, \quad R(w) = \max_i w_i - \min_i w_i, \quad (1)$$

$$\text{Var}(w) = \frac{1}{N} \sum_{i=1}^N (w_i - \bar{w})^2. \quad (2)$$

Range is understandable and easy to model linearly, but ignores changes between the extremes.

Jain's fairness index is

$$J(w) = \frac{(\sum_i w_i)^2}{N \sum_i w_i^2}. \quad (3)$$

For a nonzero nonnegative allocation, $1/N \leq J(w) \leq 1$, with one indicating equality [3]. It is a descriptive metric, not a MILP objective. The uncorrected Gini coefficient is

$$G(w) = \frac{\sum_{i=1}^N \sum_{j=1}^N |w_i - w_j|}{2N \sum_i w_i}. \quad (4)$$

Higher Gini indicates greater inequality. Both measures include all pharmacies, including zero H24 counts. The maximum finite- N Gini is $(N-1)/N$. The code uses Jain = 1 and undefined Gini for an all-zero allocation; no reported schedule has zero aggregate workload.

3.3 Baseline results and plot

Across the 53 source blocks, historical weekly staffing ranges from 17 to 18 pharmacies, with exactly three H24 pharmacies in every block. The table summarizes workload counts over the full horizon.

Table 2: Notebook 01: descriptive historical baseline.

Measure	Total workload	H24 workload
Assignments	931	159
Minimum	6	0
Maximum	8	3
Range	2	3
Jain	0.995456618	0.867242976
Gini	0.033306262	0.176100629

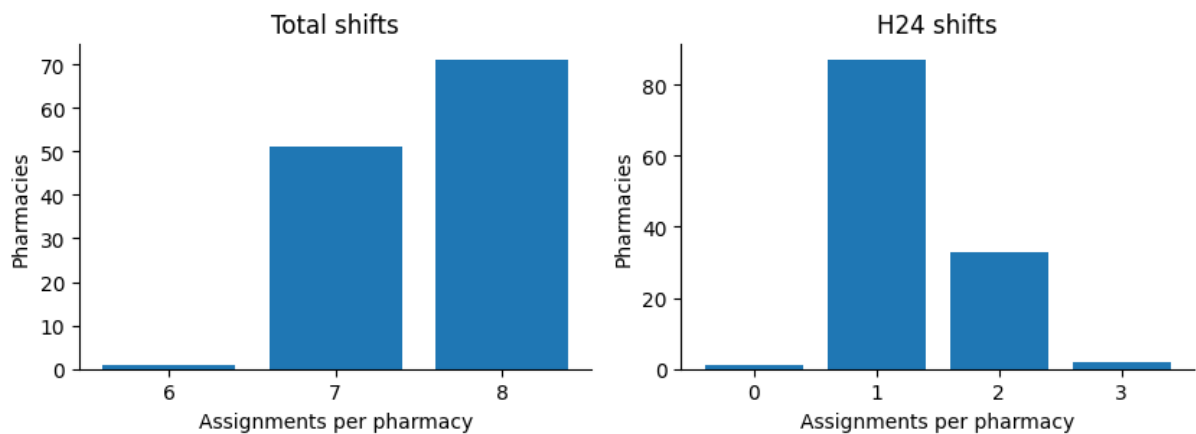


Figure 2: Notebook 01 output: historical total and H24 workload distributions. Counts include all 123 pharmacies.

The historical total-workload distribution contains one pharmacy with six assignments, 51 with seven and 71 with eight. H24 counts are zero for one pharmacy, one for 87, two for 33 and three for two. The plot displays these exact counts. All 11 repeated-pattern comparisons match the full pharmacy/location/shift/access signature. The inferred groups have sizes 17, 18, 18, 17, 18, 18 and 17.

3.4 Exceptions and interpretation

P0076 contributes the only observed cross-group row: it appears in group 7 at $nT=21$ although its dominant membership is group 5. This row is retained in the historical comparator and passed as evidence to notebook 02. There are no within-pharmacy changes in P/C label. One 53-block calendar provides no empirical test of stability in another year; the observed extrema are not independently validated standards.

3.5 Final outputs and hand-off

`baseline_evidence.json`, `baseline_evidence.csv`, `baseline_pharmacy_profile.csv`, and `stage_01_manifest.json`. These files are supplied in `notebook_01/outputs/`. The executed notebook retains its displayed tables and figures.

4 Notebook 02: constraint specification

4.1 Purpose and inputs

Notebook 02 consumes the canonical schedule and notebook 01 baseline evidence. It re-derives the counts and group maps, checks that they agree, and creates the single model contract used by notebooks 03, 04 and 05. This is where observations are converted into explicit modeling policies.

4.2 Observations, derived values and policy choices

Table 3: Provenance of observations and modeling policies.

Parameter or choice	Value / policy	Evidence status
Weekly staffing floor	17	Historical minimum used as an unvalidated proxy
Horizon workload ceiling	8	Historical maximum used as an unvalidated proxy
Weekly H24 count	Exactly 3	Observed constant retained by modeling policy
Primary-group eligibility	Dominant historical group	Observed association converted into an eligibility assumption
Shared nT decisions	All occurrences identical	Observed recurrence retained as policy
H24 cycle counts per pharmacy	1-2 in 17-member groups; exactly 1 in 18-member groups	Normative equal-count allocation derived from 18 cycle slots
Recommended objective	Total range, H24 range, assignment count	Normative priority
Historical changes	Lower-priority tie-breakers	Normative continuity preference

The cycle bounds deserve particular emphasis: six patterns per group times three H24 duties gives 18 H24 slots. Allocating those as evenly as possible yields floor($18/\text{group size}$) and ceiling($18/\text{group size}$). This is a hard equal-count fairness policy. Arithmetic derives its values, but does not make the policy an observed fact. Because the CSV has no capability variable, service type is modeled as reassignable among the primary-group members. This is an explicit symmetry assumption used for count-based optimization, not an additional capability observation.

Counts also omit differences in holidays, call duty, demand, staff size, costs, income and actual duration. A burden-adjusted objective would need weights unsupported by this dataset. Accordingly, no such weights are invented; total and H24 counts are analyzed separately.

4.3 Why the policies are adopted

The distinction between data and design is not left to implicit code behavior. Notebook 02 exports the justifications below in `model_specification.json` and `policy_justifications.csv`. Notebook 03 consumes its scenario priorities; notebook 04 checks its recovery policy; notebook

05 consumes the same recommendation order. Thus a downstream notebook does not silently introduce a different objective or reinterpret a historical candidate as authorization.

Table 4: Justification of every model and recovery policy.

Policy	Justification
staffing floor	Retain the minimum observed active-pharmacy count to avoid lowering the source calendar's minimum staffing.
workload ceiling	Do not exceed the largest observed pharmacy workload in the same 53-block horizon.
weekly h24	Preserve the uniform H24 count measured in every source block.
primary group eligibility	Use each pharmacy's dominant observed group to preserve the highly recurrent rotation; the observed exception is retained separately.
all group members can receive h24 in model	The supplied file has no capability variable. Treat service type as a reassignable decision within the inferred group; this is a symmetry assumption, not a capability observation.
h24 cycle bounds	Allocate each group's cycle H24 slots as evenly as integer counts permit, without introducing unobserved burden weights.
same nt	Reuse one decision per nT because equal nT patterns repeat exactly in the source.
fairness first	Prioritize total-count disparity, then H24 disparity, then assignment count as the stated ethical objective of this scenario.
historical stability	Break higher-objective ties by minimizing changed calendar active/H24 decisions, preserving observed continuity.
absence experiment	Enumerate one hypothetical absent pharmacy for each eligible pharmacy/nT pair; no absence log or probability is claimed.
recovery policy	Preserve staffing, cycle bounds, ceiling and attained H24 bounds; allow total range to worsen after a disruption so recovery measures feasibility rather than preservation of every optimal objective.
cross group boundary	Only exact observed cross-group occurrences suggest candidates; no authorization value is invented.

4.4 Structural group-size bound

Under the current membership, groups 1, 4 and 7 have 17 members. With a staffing floor of 17, removing any member leaves only 16 available pharmacies, so same-group recovery cannot meet staffing. The other four groups have 18 members. A spare member alone is not sufficient proof of recovery: annual limits and H24 constraints must also pass, as checked experimentally.

The counting proof requires precise assumptions. Making every one of seven groups robust to one absence under the floor requires at least $7 \times 18 = 126$ pharmacies, more than the available 123. If every group must first contain at least 17 pharmacies, allocating that minimum uses $7 \times 17 = 119$ pharmacies and leaves four. Those four can provide a spare to at most four groups; at least three groups therefore have exactly 17 members. The observed 17/18 distribution attains that bound and is maximally balanced by group size.

The identities 1, 4 and 7 are not fixed by mathematics. A repartition can move which groups are fragile. If groups below the staffing floor were allowed, even the claim of at least three small groups would fail: six groups of 18 and one of 15 also total 123, but the 15-member group cannot staff an ordinary week. The proof excludes that infeasible repartition by its explicit minimum-size assumption.

The proof does not establish that observed membership minimizes affected calendar weeks, weighted absence risk, geographic harm or workload disruption. Those are different optimization objectives. In particular, nT patterns occur with different multiplicities over the 53-block horizon.

4.5 Historical cross-group evidence

The P0076 event is stored with its exact nT and observed secondary group. It supports a candidate for a hypothetical exception in notebook 04, not a general expansion of eligibility. The base optimizer retains dominant-group eligibility. No actual absence, capability or approval variable is invented.

4.6 Final outputs and hand-off

`model_specification.json`, `parameter_provenance.csv`, `secondary_eligibility_evidence.csv`, `policy_justifications.csv`, and `stage_02_manifest.json`. These files are supplied in `notebook_02/outputs/`. The executed notebook retains its displayed tables and figures.

5 Notebook 03: optimization and fairness evaluation

5.1 Purpose and inputs

Notebook 03 reads the canonical schedule and the justified model specification from notebook 02. It verifies all upstream stages, constructs the MILP below, solves the three prescribed scenario orders, materializes the resulting calendar rows and recomputes their metrics. It does not introduce new pharmacy attributes or a new fairness priority.

5.2 Mathematical model and lexicographic solution

Let I be the set of pharmacies, T the set of 42 distinct patterns, $g(t)$ the structural group of pattern t , g_i the dominant observed group of pharmacy i , and m_t the number of calendar occurrences of t . The eligibility set is

$$E = \{(i, t) \in I \times T : g_i = g(t)\}. \quad (5)$$

For $(i, t) \in E$, let $y_{it} \in \{0, 1\}$ indicate active duty and $h_{it} \in \{0, 1\}$ indicate H24 duty. Noneligible pairs are excluded from the decision-variable set.

The horizon total workload, horizon H24 workload and unweighted H24 cycle workload are

$$W_i = \sum_{t:(i,t) \in E} m_t y_{it}, \quad H_i = \sum_{t:(i,t) \in E} m_t h_{it}, \quad C_i = \sum_{t:(i,t) \in E} h_{it}. \quad (6)$$

The cycle quantity counts distinct patterns, while the horizon quantities count calendar occurrences. The complete hard-constraint system is

$$\sum_{i:(i,t) \in E} y_{it} \geq s \quad \forall t \in T, \quad (7)$$

$$\sum_{i:(i,t) \in E} h_{it} = q \quad \forall t \in T, \quad (8)$$

$$h_{it} \leq y_{it} \quad \forall (i, t) \in E, \quad (9)$$

$$W_i \leq U \quad \forall i \in I, \quad (10)$$

$$a_{g_i} \leq C_i \leq b_{g_i} \quad \forall i \in I, \quad (11)$$

$$W^{\min} \leq W_i \leq W^{\max} \quad \forall i \in I, \quad (12)$$

$$H^{\min} \leq H_i \leq H^{\max} \quad \forall i \in I. \quad (13)$$

Here $s = 17$, $q = 3$, and $U = 8$ are the declared data-derived values. For a group of size n_g with $k_g = 6$ distinct patterns,

$$a_g = \left\lfloor \frac{k_g q}{n_g} \right\rfloor, \quad b_g = \left\lceil \frac{k_g q}{n_g} \right\rceil. \quad (14)$$

The range objectives are $R_W = W^{\max} - W^{\min}$ and $R_H = H^{\max} - H^{\min}$; assignment minimization uses $A = \sum_i W_i$.

The variables y and h are binary. The four auxiliary extrema are continuous and bounded between zero and the horizon ceiling. Pattern-level variables enforce identical repeated nT assignments by construction. Location and access mode are carried from observed pharmacy metadata and checked in the materialized rows; neither is optimized.

The mixed-integer problem is solved through SciPy’s milp interface to HiGHS. Each lexicographic stage must terminate successfully at zero reported relative MIP gap; bounds, integrality and linear-constraint residuals are checked before the objective is locked. Solver objectives and dual bounds are exported in `solver_evidence.json`. SciPy documents the MILP status and gap fields used for these checks. [4]

The scenario priority orders are:

- Efficiency-first: assignments, total range, H24 range, active changes, H24 changes.
- Fairness-aware: total range, H24 range, assignments, active changes, H24 changes.
- Full-group: fix every eligible y_{it} to one, then minimize H24 range and H24 changes.

Changes are weighted by calendar occurrences. Their linear coefficient is m_t minus twice the number of historical occurrences of that decision; a constant can be omitted without changing the minimizing solution. The unavoidable removal of an ineligible historical cross-group row is another constant. These tie-breakers minimize changed active/H24 decisions, not the number of distinct changed patterns. Equal objective values can therefore allow different detailed schedules across solver versions.

The historical schedule is a descriptive comparator. It contains a cross-group exception and need not satisfy the revised eligibility and cycle-balance policy; it is not presented as a feasible point of the same model.

5.3 Final optimized results

Table 5: Notebook 03: assignment totals and workload ranges.

Scenario	Assignments	Total range	H24 range
Historical	931	2	3
Efficiency-first	901	3	1
Fairness-aware	923	1	1
Full-group	931	1	1

The fairness-aware schedule removes 8 assignments (0.86% of history) and uses 22 more assignments than efficiency-first. It narrows total workload from 6-8 to 7-8 assignments and H24 workload from 0-3 to 1-2 assignments per pharmacy. Full-group matches those ranges but uses eight more assignments. It is dominated under the selected three metrics, yet has better total Jain and Gini than fairness-aware. Thus the preferred scenario depends on the specific fairness objective, not on a universal ordering of fairness.

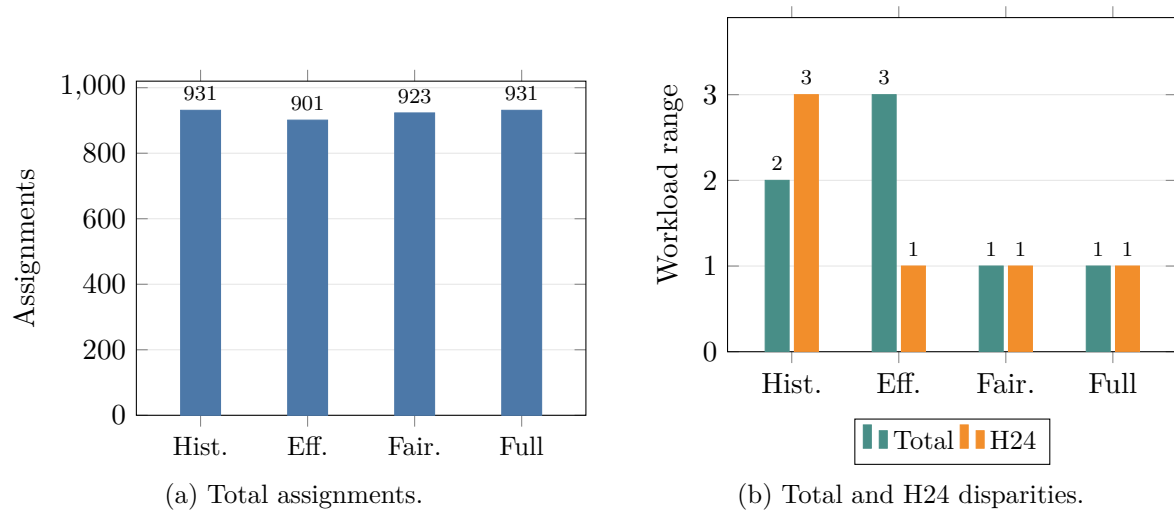


Figure 3: Fairness and efficiency trade-off across the observed and optimized schedules. Both panels start at zero.

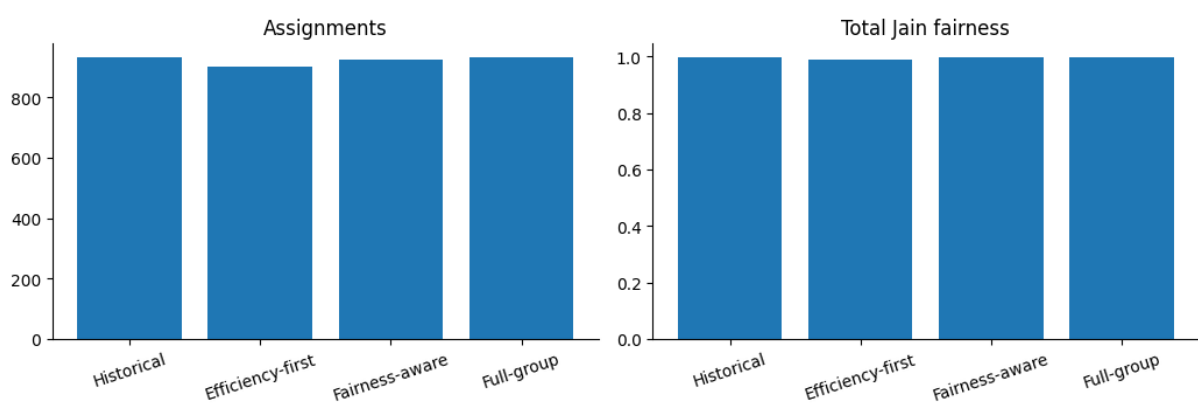


Figure 4: Notebook 03 output: scenario assignment counts and total Jain fairness. Precise fairness values appear in the adjacent tables.

The second plot is the actual output of notebook 03. Its full-scale Jain axis makes the small total-Jain differences visually subtle; the numerical tables provide the precision needed to compare them.

Table 6: Historical and fairness-aware workload metrics.

Measure	Historical	Fairness-aware
Assignments	931	923
Total workload range	2	1
H24 workload range	3	1
Total Jain	0.995456618	0.995580284
Total Gini	0.033306262	0.033313074
Total population variance	0.261484566	0.249983475
H24 Jain	0.867242976	0.889768768
H24 Gini	0.176100629	0.160147262

Total Jain increases slightly. Total Gini increases from 0.033306262 to 0.033313074, a small worsening of 0.000006812; both round to 0.0333 at four decimals. It would be incorrect to say that all fairness indices improve. The strongest positive results are the reduced total and H24 ranges and the improved H24 distribution.

5.4 All scenario fairness statistics

Table 7: Complete scenario count and fairness statistics.

Scenario	Total min	Total max	H24 min	H24 max	Total Jain	Total Gini	H24 Jain	H24 Gini
Historical	6	8	0	3	0.995457	0.033306	0.867243	0.176101
Efficiency-first	5	8	1	2	0.988469	0.054050	0.889769	0.160147
Fairness-aware	7	8	1	2	0.995580	0.033313	0.889769	0.160147
Full-group	7	8	1	2	0.995738	0.032398	0.889769	0.160147

5.5 Workload distributions

Table 8: Workload distributions, including zero H24 counts.

Schedule	Workload type	Assignments per pharmacy	Pharmacies
Historical	Total	6	1
Historical	Total	7	51
Historical	Total	8	71
Historical	H24	0	1
Historical	H24	1	87
Historical	H24	2	33
Historical	H24	3	2
Fairness-aware	Total	7	61
Fairness-aware	Total	8	62
Fairness-aware	H24	1	87
Fairness-aware	H24	2	36

These distributions make the range and inequality calculations auditable from counts of pharmacies rather than only aggregate indices. All N pharmacies enter each distribution; zeros are included for historical H24.

5.6 Changes measured at calendar level

Table 9: Historical changes at calendar-decision level.

Scenario	Active decision changes	H24 decision changes
Historical	0	0
Efficiency-first	32	6
Fairness-aware	10	6
Full-group	2	6

A changed active or H24 decision means an added or removed pharmacy-week binary decision. Replacing one pharmacy with another counts as two changes, and recurring nT patterns contribute once for each affected calendar occurrence. These counts therefore differ from the number of distinct changed patterns.

5.7 Independent bounds explaining the optima

5.7.1 Assignment-minimizing scenario

Every one of 53 blocks requires at least 17 active pharmacies, so every feasible schedule has at least $53 \times 17 = 901$ assignments. Efficiency-first attains 901, proving its assignment total is

minimal under the model. The subsequent range objectives are solved after fixing this assignment optimum. Their reported values are supported by the corresponding MILP certificates.

5.7.2 Fairness-first total range

Groups 1 and 4 each have 17 members and eight calendar occurrences. Every member must therefore work eight times because the staffing floor equals the entire group size. Groups 5 and 6 have only seven occurrences each, so their members cannot exceed seven assignments under primary-group eligibility. Consequently, total range zero is impossible and range at least one is required. The fairness-aware schedule attains range one.

Once range one is fixed, every pharmacy must receive at least seven assignments. Groups 1 and 4 contribute $34 \times 8 = 272$. Groups 2 and 3 each contain 18 pharmacies and occur eight times through six distinct patterns: two occur twice and four once. Removing a pharmacy from a repeated pattern would reduce its total from eight to at most six, violating the seven-assignment minimum implied by range one. Only the four singleton patterns may omit one member each; the staffing floor prevents omitting more than one in a pattern. Each of these groups therefore contributes at least $18 \times 8 - 4 = 140$ assignments. Groups 5 and 6 together contain 36 pharmacies, and group 7 contains 17; all these 53 pharmacies must work at least seven times. Hence the fairness-first assignment lower bound is $272 + 2 \times 140 + 53 \times 7 = 923$. The exported fairness-aware schedule attains it.

5.7.3 H24 range

Exactly three H24 assignments per block gives $3 \times 53 = 159$ horizon H24 assignments. Since 159 is not divisible by 123, equal integer H24 workloads across all pharmacies are impossible. H24 range is therefore at least one; the optimized scenarios attain one. The independent counting bounds explain the leading objectives, while solver evidence covers the full lexicographic chains and their historical-stability tie-breakers.

5.7.4 Meaning of the stability objective

For an eligible binary decision with m calendar occurrences and k historical presences, its Hamming-distance contribution is $k + (m - 2k) \times \text{decision}$. The constant k can be omitted by the solver. Negative signed objective values in `solver_evidence.json` are therefore valid and must not be read as negative numbers of changes. The report's calendar-change table computes the actual symmetric differences, including constants and the historical cross-group exception.

5.8 Final outputs and hand-off

`optimized_schedule_fairness_aware.csv`, `optimized_schedule_efficiency_first.csv`, `optimized_schedule_full_group.csv`, `optimization_scenario_summary.csv`, `applied_model_specification.json`, `solver_evidence.json`, and `stage_03_manifest.json`. These files are supplied in `notebook_03/outputs/`. The executed notebook retains its displayed tables and figures.

6 Notebook 04: hypothetical absences and substitutions

6.1 Purpose, inputs and experimental unit

Notebook 04 takes the selected fairness-aware schedule, the canonical historical metadata and the notebook 02 contract. The optimized schedule and all upstream hashes are checked before any experiment. Each event is a hypothetical unavailable pharmacy at one distinct nT pattern, affecting every occurrence of that pattern.

The experiment considers 738 eligible pharmacy-pattern pairs: each of 123 pharmacies has six primary-group patterns. 8 cases involve an inactive pharmacy and require no response; they are excluded from the recovery-rate denominator. The 730 scheduled cases are evaluated one at a time from the same base schedule, not cumulatively.

An absence removes a pharmacy from every calendar occurrence of its nT. When staffing falls below the floor, the policy considers a same-group spare without exceeding the horizon workload ceiling. If the absent pharmacy was H24, a small MILP reassigns H24 across all six patterns in that group. It preserves the H24 cycle bounds and the base schedule’s annual H24 minimum and maximum. The optimal total-workload range is not locked during recovery, so disruptions can worsen total-count fairness.

Every reported successful response is expanded to calendar rows and checked for week metadata, absence compliance, unique pharmacy-week assignments, group eligibility, staffing, H24 totals, cycle bounds, horizon ceiling and repeated-pattern identity. The exported response CSV contains H24 changes, and `recovery_pattern_plans.json` records H24 sets across the affected group. A recovery can require changes at patterns other than the absent one, so it is not a simple last-minute one-week substitution and may be unsuitable once earlier duties have already occurred.

Table 10: Scheduled pattern-level absence cases by rotation group.

Group	Size	Scheduled cases	Within group	Candidate / approval	No candidate
1	17	102	0	0	102
2	18	104	104	0	0
3	18	104	104	0	0
4	17	102	0	0	102
5	18	108	108	0	0
6	18	108	108	0	0
7	17	102	0	17	85

The same-group modeled recovery rate is $424/730 = 58.1\%$. All 306 remaining cases (41.9%) are outside same-group recovery. Human intervention here means choosing whether to authorize a cross-group substitution or explicitly change a modeled constraint; no such choice is made automatically. Of these, 17 have a historical candidate and 289 have no pre-filled candidate. The authorization template includes all 306, with approval fields left blank.

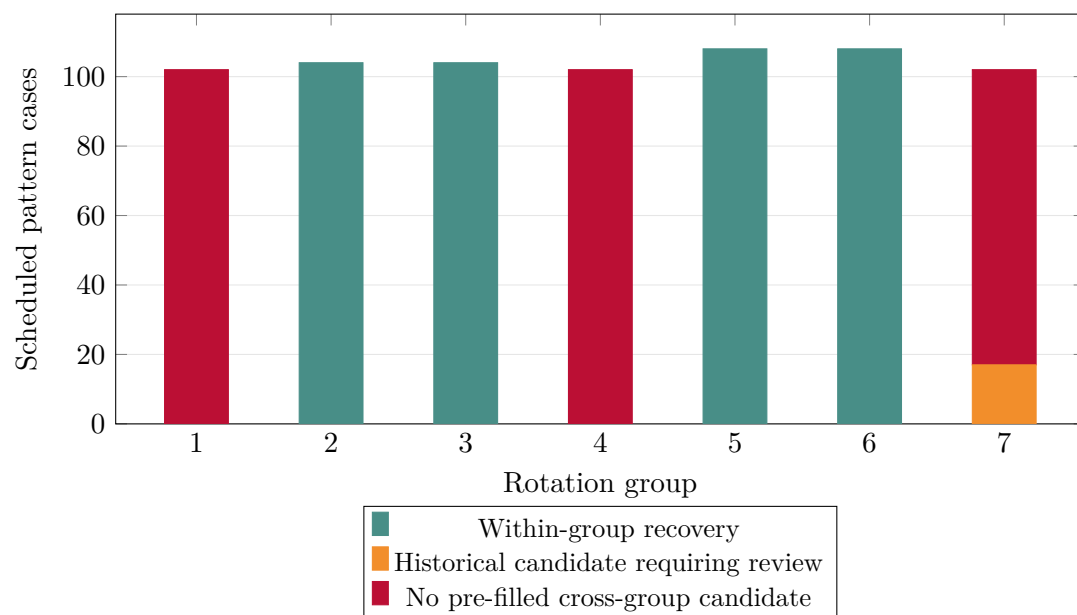


Figure 5: All scheduled hypothetical absence cases. Candidate cases remain separate from within-group recoveries.

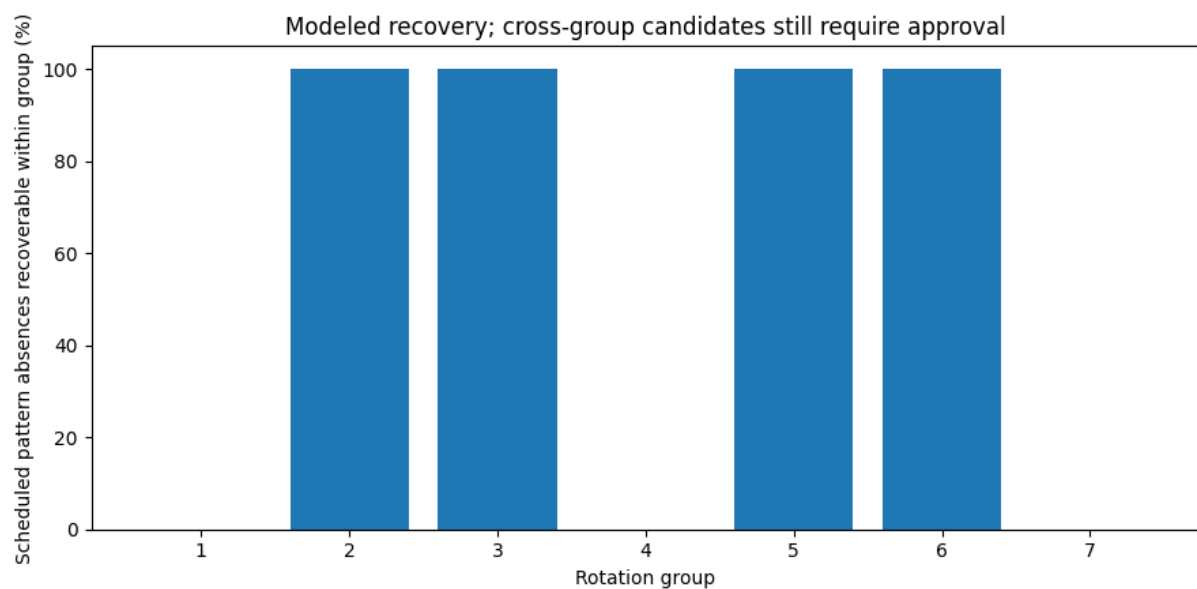


Figure 6: Notebook 04 output: within-group recovery rates among scheduled pattern-level absences. A zero bar means no same-group recovery under the stated model.

The observed exception is P0076 at $nT=21$ in group 7, despite a dominant group of 5. The source documents the occurrence, not its administrative reason, current availability, or authorization to replace every absent group-7 member. For the 17 corresponding cases, the candidate is added as STANDARD; when H24 is lost, the group's H24 re-optimization supplies it. 17 of these candidate schedules pass all represented constraints using the exact historical eligibility exception. This modeled check still cannot verify legal eligibility, availability, geography or authority approval. They are never counted as automatic recoveries. Within this academic project, leaving these approval fields blank is the completed representation of that boundary; filling them with invented approvals would be incorrect.

All 126 of the 126 attempted H24 recovery models are feasible. Some belong to cases where staffing still fails. H24-only feasibility is therefore not full schedule feasibility, and this result does not establish that H24 will never be a bottleneck in other absence patterns or operational settings.

The denominator is an unweighted set of pattern cases, not an estimated probability distribution of absences. A 58.1% case recovery rate is not a forecast of the fraction of real disruptions that will be manageable. Week-level, multi-absence, correlated-disruption and demand-change experiments remain unimplemented.

6.2 What requires a person

The staffing obstacle is concrete: a size-17 group cannot supply 17 active pharmacies after one member is removed. A cross-group assignment changes the primary-group eligibility boundary, while reducing the staffing floor changes the declared constraint. Those policy decisions are deliberately left to a person. The 17 pre-filled candidates are limited to the exact historical P0076/ $nT=21$ exception; the other 289 cases carry no evidence-based substitute suggestion. A blank authorization template is the completed experimental output, not an incomplete data-collection step.

6.3 All scheduled-absence response categories

Table 11: All scheduled-absence response categories.

Response category	Cases
h24 reoptimization within group	48
historical candidate requires approval	17
manual cross group decision required	289
same group active replacement	112
same group active replacement and h24 reoptimization	24
spare capacity absorbs absence	240

6.4 Worked response example

Consider the simulated absence of P0030 at $nT=9$, group 2. This pharmacy is active and H24 in the selected schedule. The same-group active replacement is P0021. The saved H24 changes are: ADD:P0018@ $nT=9$; REMOVE:P0030@ $nT=9$; REMOVE:P0018@ $nT=37$; ADD:P0030@ $nT=37$. The complete revised schedule passes staffing, H24, horizon workload, cycle-balance, absence, eligibility and repeated-pattern checks. This is an illustrative row selected from the exported response plan, not an invented event or an observation of an actual absence.

6.5 Recovery constraints and invariants

The activity update is local to the absent nT , but H24 may be reassigned across the affected group's six patterns. For each candidate H24 assignment, a binary decision is allowed only if that pharmacy remains active at that pattern. Every group pattern retains three H24 pharmacies; each pharmacy retains its original cycle lower/upper bound and its base optimized horizon H24 interval of 1-2. Unaffected groups remain unchanged. The activity count must stay at least 17 and no pharmacy may exceed eight total assignments. The unavailable pharmacy is absent from every occurrence of the tested nT . Total workload range one is not imposed during recovery.

No same-group spare exists in a size-17 group. In a size-18 group, there is at most one spare when only 17 are active, so candidate enumeration is exhaustive for this dataset. The exported successful schedules and the reported same-group failures are independently rechecked. Cross-group cases remain separate, even when a historically evidenced candidate produces a feasible modeled schedule.

6.6 Final outputs and hand-off

`single_absence_response_plan.csv`, `robustness_summary.csv`, `authorized_substitutions_template.csv`, `secondary_eligibility_candidates.csv`, `robustness_evidence.json`, `recovery_pattern_plans.json`, and `stage_04_manifest.json`. These files are supplied in `notebook_04/outputs/`. The executed notebook retains its displayed tables and figures.

7 Notebook 05: explanation and final comparison

7.1 Purpose and inputs

Notebook 05 combines the saved schedules, scenario metrics, model specification and robustness evidence from notebooks 00-04. It verifies their explicit hashes, recomputes the main schedule metrics, compares full pattern signatures and applies the recommendation order already specified in notebook 02. It is the synthesis stage, not a second optimization model.

7.2 Final scenario comparison and recommendation

Table 12: Final scenario comparison under the declared three-metric decision rule.

Scenario	Assignments	Total range	H24 range	Nondominated*	Selected
Fairness-aware	923	1	1	True	True
Full-group	931	1	1	False	False
Historical	931	2	3	False	False
Efficiency-first	901	3	1	True	False

*The table's nondominance labels compare the four displayed candidates over assignments, total range and H24 range. They do not enumerate the full feasible frontier. For the lexicographically optimized scenarios, optimality is separately supported by the solver evidence. Dominance uses only the declared comparison metrics and is not a claim about every ethical consideration.

The selected fairness-aware scenario minimizes total range, then H24 range, then assignment count among the displayed candidates. Efficiency-first is also nondominated under those three metrics. Full-group matches both selected ranges but uses eight more assignments. Its better descriptive total Jain and Gini illustrate that this recommendation depends on the declared comparison rule.

7.3 Complete pattern explanations

There are 11 changed full patterns and 31 unchanged patterns in this executed solution. 9 patterns change active membership and 3 change H24 membership; 2 have H24-only changes. `nt_change_explanation.csv` records both types, including added and removed H24 IDs. A count based only on active pharmacies misses the H24-only changes. Patterns should not be called overstaffed merely because they exceed the historical-minimum proxy.

Table 13: Every complete nT pattern changed by the selected solution.

nT	Before	After	Active added	Active removed	H24 added	H24 removed
9	18	17	-	P0021	-	-
10	18	17	-	P0049	-	-
16	18	17	-	P0020	-	-
17	18	17	-	P0045	-	-
21	17	17	P0111	P0076	P0111	P0076
22	17	17	-	-	P0009	P0003
23	18	17	-	P0029	-	-
24	18	17	-	P0039	-	-
25	17	17	-	-	P0057	P0067
30	18	17	-	P0034	-	-
31	18	17	-	P0037	-	-

The nT=21 change replaces the historical cross-group row by a primary-group member under the stated eligibility policy. Patterns 22 and 25 change H24 assignments without changing active membership. The other eight active-set changes each remove one active assignment from a singleton pattern while preserving the staffing floor and total range. No new pharmacy identity is created by optimization.

7.4 Ethical interpretation and oversight

Duty scheduling affects pharmacy organizations, workers, citizens and scheduling authorities. Fewer assignments can reduce measured workload but also remove redundancy, income opportunities or service availability. Equal pharmacy counts do not establish equal public access. The model’s preferred outcome therefore depends on the objective chosen by the scheduling authority.

The European Commission’s Trustworthy AI guidelines identify human oversight, technical robustness, privacy and data governance, transparency, fairness and accountability among their requirements. These provide an ethical framing for this case study; passing the computational checks is not certification against those guidelines. [1]

No protected-attribute, ownership, workforce, revenue or preference data are included. The project measures count-based allocation fairness among modeled pharmacies. It does not establish demographic parity, equal opportunity, legal compliance or absence of discrimination. Pseudonymizing names also does not remove inequities potentially embedded in historical group membership.

Implemented explanations comprise parameter provenance, explicit model constraints, lexicographic objective order, scenario comparisons, complete pattern differences and per-event recovery details or constraint failures. They show what changes and why a stated requirement blocks a response. They do not uniquely explain why one pharmacy was selected among all alternative optima.

For example, a scheduled absence in a 17-member group leaves 16 members against the required 17. This explains a staffing failure under the assumed floor. It is not evidence that 17 is a socially sufficient service level. Similarly, a suggested cross-group candidate is historical evidence, not a permission decision.

Counterfactual re-solving (forcing a particular alternative pharmacy) and formal irreducible infeasible subsystem extraction are proposed extensions. They are not implemented or claimed as completed features. A future counterfactual service should state whether an alternative is infeasible, worsens a higher-priority objective, or is equally optimal.

Human oversight is represented by separating computed proposals from authorization. The project intentionally leaves approval fields blank and never turns a historical occurrence into an approval record. This is a complete decision-support design for the supplied evidence, rather than an unfinished effort to obtain unprovided real-world approvals. If the prototype were ever used beyond this coursework scope, responsibility for actual scheduling decisions would remain with the relevant authority.

7.5 Final validation and reproducibility

Table 14: Validation within the agreed dataset-only scope.

Component	Evidence	Status
Data and implementation lineage	Stage manifests and explicit canonical/specification/optimized/robustness links verified	PASS
All scenario schedules	All exported schedule constraints and summary metrics checked	PASS
Pattern-level recovery	Every reported same-group success and historical candidate materialized and checked	PASS
Cross-group boundary	306 simulated cases need a cross-group decision; no approval records are invented	DOCUMENTED MODEL OUTCOME
Group bound	PROVEN: no repartition of 123 pharmacies into 7 groups can raise every group above 17. If all groups must contain at least 17 members, the surplus 4 can give a spare to at most that many groups, so at least 3 groups have zero slack. The observed grouping attains this bound. Group identities are not mathematically fixed; this does not prove minimum numbers of affected weeks or absence scenarios.	PROVEN UNDER STATED ASSUMPTIONS
Staffing and workload thresholds	Derived from source minima/maxima and retained by explicit continuity policy	JUSTIFIED MODEL CHOICE
Calendar boundary	Final block ends 2027-01-03, six elapsed days after Monday start; source preserved and limitation documented	CAVEAT
Dataset-only scope	No geography, actual availability, demand, preferences or external rules added; excluded by project scope	SCOPE SATISFIED
Recovery scope	Pattern-level, one absence at a time; H24 moves may affect other patterns, total fairness is not locked	CAVEAT

Validation has two layers. Shared cross-notebook checks verify the declared constraints, materialized rows, scenario metrics and artifact lineage. A separate audit independently reconstructs every source row, computes direct Jain and pairwise Gini, checks all three saved optimized schedules, reconstructs all 738 recovery plans and tests rejection of corrupted schedules and stale files. The audit also checks an existing-ID mapping extension on a synthetic fixture, without adding those fixture records to the analysis.

The notebook folders contain the executed notebook, a frozen `inputs/` bundle, an `outputs/` directory and a `run_notebook.py` launcher. Upstream notebook source snapshots under `inputs/provenance/` are required by the existing lineage checks; they are not extra analyses. Because the original cells use relative `data/` and `output/` paths, the launcher assembles those paths in a temporary working directory, executes the unchanged cells and copies only the current stage's artifacts back to its output folder. Each folder can be run independently from its supplied snapshot. The root launcher runs all stages in order and refreshes downstream snapshots from the results it has just produced.

The report is an editorial reorganization of notebook 05's generated evidence narrative. Tables and figures retain the verified output values; the chapter order and explanations are supplied by the report builder. The original generated narrative remains in notebook 05's outputs so that its code is unchanged. The typeset report and its reorganized Markdown are in the separate report folder.

Commands from the archive root:

```
1 python -m pip install -r requirements.txt
2 python run_project.py
3 python audit_project.py
4 python report/render_report.py
```

To rerun one stage independently, use `pythonnotebook_03/run_notebook.py`, replacing 03 with the desired stage number. Python 3.12 and the pinned dependencies were used. PDF compilation needs a standard TeX Live installation. Alternatively, upload the contents of `report/` to Overleaf and compile `final_report.tex`. Name and email remain editable in `report/metadata.tex`.

Presentation validation compares the executable syntax trees and all non-comment tokens with the previously verified notebooks. The only notebook-source changes are removal of comment tokens and replacement of Markdown prose with one title before each code cell. Fresh execution refreshes the source manifests after those presentation edits. No modeling or computational statement is changed.

7.6 Interpretation within the agreed scope

The project is a completed analysis of a single calendar under an explicit count-based model. Four boundaries matter for interpreting its results:

1. **Historical rather than externally validated thresholds.** The minimum staffing count and maximum workload are genuine CSV extrema retained by policy. Their values are justified for historical continuity, not claimed to be universal standards.
2. **Equality of counts.** The model optimizes total and H24 workload ranges. It neither estimates absent burden weights nor treats improvement in one index as improvement in all indices.
3. **Hypothetical pattern-level disruptions.** The absence scenarios are generated from observed entities and patterns; they are not records of actual absences or estimates of absence probabilities. Repeated occurrences and cross-pattern H24 moves remain explicit.
4. **Single-source evidence.** Results describe this 53-block calendar. The final six-day boundary is preserved. No claim of another year's behavior, actual authorization or unrecorded information is made.

Geography, actual availability, demand, preferences and external rules are excluded from the project by design. Their absence does not make the dataset-only submission incomplete. Similarly, 306 cases outside automatic within-group recovery are a substantive experimental result, not a requirement to invent substitute approvals. Broader operational extensions are optional future research, not conditions for completing this project.

7.7 Conclusion

The optimization formulation is appropriate for the supplied pharmacy-rotation data and the stated count-based fairness objective. It preserves the modeled historical rotation structure and staffing/H24 constraints while reducing total assignments from 931 to 923, total range from 2 to 1 and H24 range from 3 to 1.

The improvement is precise rather than universal: total Jain improves slightly, total Gini worsens slightly, and full-group is better on those two descriptive total-count indices while using more assignments. There are 11 changed complete patterns, including 2 H24-only changes. The experiment finds 424 within-group recoveries among 730 scheduled pattern cases and 306 cases requiring a cross-group decision; 17 have a historically supported, model-feasible candidate without an invented approval.

Every numerical result is either reconstructed from the supplied CSV, deterministically derived from it, computed by the stated optimization, or produced by the explicitly defined hypothetical experiment. Normative and computational choices are documented and justified. Within this dataset-only scope, the computational work and report constitute a complete, reproducible submission.

7.8 Final outputs and hand-off

`final_project_report.md`, `nt_change_explanation.csv`, `scenario_pareto_comparison.csv`, `final_validation_checklist.csv`, `final_project_evidence.json`, and `stage_05_manifest.json`. These files are supplied in `notebook_05/outputs/`. The executed notebook retains its displayed tables and figures.

A Artifact lineage and archive guide

Each stage records hashes of relevant inputs, outputs and implementation modules; downstream stages check those manifests before use. Notebook 05 additionally checks the explicit canonical, applied-specification, optimized-schedule and robustness links. Hashes detect accidental stale or altered artifacts relative to recorded manifests; they are not authentication, external validation or proof that a formula is correct.

Table 15: Dependency trace from the source CSV to the final report.

Stage	Direct input	Derivation or decision	Output used downstream
00	Original TURNI CSV	Parse sections/dates, normalize and pseudonymize, infer and test recurring group structure	Canonical schedule and observed profile
01	Canonical schedule	Count weekly and pharmacy workloads; measure H24, P/C, groups, repeats and exceptions	Baseline evidence and pharmacy profile
02	Canonical schedule and baseline evidence	Re-derive values; justify policies; state mathematical group bound	Model specification, parameter and policy provenance
03	Canonical schedule and model specification	Solve declared lexicographic scenarios and materialize every assignment	Three schedules, all scenario metrics, solver evidence
04	Selected schedule, canonical metadata and model specification	Generate hypothetical pharmacy/nT absences and validate recovery/candidate schedules	Response plans, group results and cross-group decision template
05	Saved schedules and verified evidence from stages 00-04	Recompute comparisons, select by the declared rule and explain changes	Final evidence, complete report text and validation checklist
Audit	Original source and saved outputs	Separate source parser, independent metric/constraint calculations and regression checks	Audit evidence and correction log
Report build	Final verified report text and saved numeric outputs	Typeset the same evidence in the supplied LaTeX style	Editable LaTeX and compiled PDF

Table 16: Artifacts supplied for reproduction and assessment.

Artifact	Purpose
notebook_00/ through notebook_05/	One executed notebook, required input snapshot, produced outputs and launcher per stage
notebook_00/inputs/data/	Original unchanged TURNI CSV
notebook_XX/inputs/provenance/	Earlier notebook source snapshots needed by lineage validation
notebook_XX/inputs/output/	Earlier stage artifacts required by the current stage and its transitive checks
notebook_XX/outputs/	Files produced by the current notebook, its manifest and extracted plot images
run_project.py	Sequential execution with fresh downstream input snapshots
audit_project.py and audit/	Independent audit launcher, verified evidence and notebook-presentation checks
requirements.txt	Pinned Python dependencies
report/Final_Report.pdf	Complete report organized by notebook
report/final_report.tex and related TeX files	Editable report in the supplied LaTeX style
report/metadata.tex	Student and submission metadata
report/final_report.md	Reorganized narrative used to build the PDF
report/inputs/ and report/figures/	Evidence snapshots and actual notebook plots used by the report
report/render_report.py and report/organize_report.py	Reproducible report assembly and compilation

References

- [1] European Commission High-Level Expert Group on AI. *Ethics Guidelines for Trustworthy AI*. 2019. <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>.
- [2] B. Phipson and G. K. Smyth. Permutation P-values Should Never Be Zero. *Statistical Applications in Genetics and Molecular Biology*, 9(1), Article 39, 2010. <https://arxiv.org/pdf/1603.05766>.
- [3] R. Jain, D.-M. Chiu and W. R. Hawe. *A Quantitative Measure of Fairness and Discrimination for Resource Allocation in Shared Computer Systems*. DEC-TR-301, 1984. <https://www.cs.e.wustl.edu/~jain/papers/fairness.htm>.
- [4] SciPy documentation. *scipy.optimize.milp*. <https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html>.